

# Power Amplifier for FM Transmitter

JOY MUKHERJI

If you have an FM transmitter in the range of 88MHz to 108MHz with power output in the range of 200mW to 500mW, here is an amplifier circuit which is capable of amplifying the existing power level to 4 to 5 watts. Use of a ground plane will further increase the range of the transmitter.

## Circuit and working

As shown in Fig. 1, the circuit is built around VHF power transistor 2SC1971(T1), four 50pF variable capacitors (VC1 through VC4) and a few other components.

As shown in the circuit diagram, Inductor L1, along with trimmer capacitors VC1 and VC2, transforms the input impedance to a very low value

required for driving transistor T1. The base of transistor T1 is connected to the ground through resistor R1. The base bias is set such that only the high peaks of the input cause the transistor to conduct. This is the basic principle behind class C amplifiers. The RFC1 choke traps the radio frequency energy at the collector of T1 but lets the DC current pass through.

The amplified signal is fed to an antenna via output matching network comprising L2, VC3 and VC4. Tune the trimmer capacitors at input and output for maximum power output. Also, spread windings of inductors L1 and L2 with your fingers until maximum output power is achieved. Tune alternately (through trimmer capacitors and inductors) until no further changes

are required. The input must not exceed 0.5W during normal operation. Mount T1 on a large heatsink. Use good-quality 50-ohm coaxial cable for input and output connections.

## Construction and testing

An actual-size, single-side PCB for the power amplifier is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, house the amplifier in a small metal box. A low-pass filter can be installed between the amplifier output and the antenna to reduce the harmonics.

## PARTS LIST

### Semiconductors:

T1 - 2SC1971 VHF power transistor

### Resistor (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 33-ohm

### Capacitors:

C1 - 1nF ceramic disk  
C2 - 100nF ceramic disk  
C3 - 100 $\mu$ F 25V electrolytic  
VC1-VC4 - 50pF trimmer capacitor

### Miscellaneous:

S1 - On/off switch  
L1 - 2 turns of 26SWG wire wound over 4mm length with 8mm dia  
L2 - 3 turns of 20SWG wire wound over 8mm length with 8mm dia  
RFC1 - 8 turns of 26SWG wire wound on TV balun core  
CON1 - 2-pin terminal connector  
Heat sink for 2SC1971  
Antenna

The author's prototype was built on a 15cm $\times$ 10cm copper-clad board. The large ground plane used is ideal for this power amplifier. The ground-plane antenna is a variant of the dipole antenna, which is designed for use with an unbalanced feed line such as coaxial cable.

Before using the amplifier, ensure that a sup-

ply voltage of 12V is available at TP1 wrt TP0 and an antenna is connected to the output. •

The author is an electronics hobbyist and a small business owner in Albany, New York, USA. His interests include designing radio-frequency circuits, among others

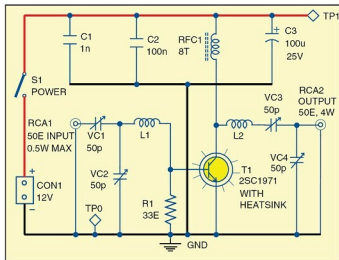


Fig. 1: Circuit diagram of power amplifier for FM transmitter

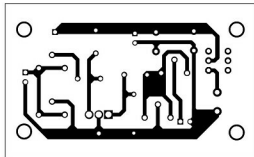


Fig. 2: Actual-size, single-side PCB for power amplifier

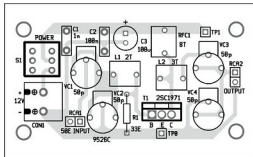


Fig. 3: Component layout for PCB